Chapter 2: Geographic Data Model

1. Introduction to GIS Data

Geographic data (or geospatial data) is unique because it combines two distinct types of information to represent real-world entities:

- **Spatial Data (Geometric Data):** Describes the absolute location, shape, size, and geometry of a feature on the Earth's surface using a coordinate reference system (e.g., Latitude/Longitude or UTM).
- **Attribute Data (Asymmetric/Descriptive Data):** Describes the non-spatial characteristics or qualitative/quantitative properties of that geographic feature (e.g., name, population, soil type, or diameter of a pipeline).

2. GIS Data Sources

Data collection is one of the most critical phases in any GIS project. Based on the literature, data sources are broadly categorized into two types:

A. Primary Data Sources (Direct Collection)

- **Digital Field Surveys:** Coordinates and feature attributes captured directly on site using total stations, digital levels, or surveying equipment.
- **Global Navigation Satellite Systems (GNSS/GPS):** Real-time spatial tracking and precise coordinate collection in the field.
- **Remote Sensing (RS):** Raw satellite imagery (e.g., Landsat, Sentinel) and aerial photography captured directly via sensors.

B. Secondary Data Sources (Derived Data)

- **Analog Maps (Hardcopy):** Paper topographic or thematic maps converted to digital format through manual digitizing tables or optical scanning.

- **Existing Digital Repositories:** National mapping agency databases, open-source clearings (like OpenStreetMap), or government shapefiles.
- **Tabular Data:** Census records, statistical sheets, or meteorological spreadsheets linked to existing spatial boundaries via a common key attribute.

3. The GIS Data Model Concept

A **Geographic Data Model** is an abstract conceptual framework that defines how real-world geographic features are represented, structured, and stored as digital data inside a computer system. It bridges the gap between complex reality and computer databases. a data model dictates how spatial relationships (topology) are maintained, how storage is optimized, and what analytical operations can be performed on the layers. The two dominant spatial data models are **Vector** and **Raster**.

4. Vector Spatial Data Model

The vector data model uses discrete coordinate geometry to represent real-world features. Space is treated as an empty canvas populated by explicit, sharply defined objects.

A. Three Fundamental Geometric Elements

- **Points:** Zero-dimensional features defined by a single coordinate pair (X, Y). Used to represent localized entities where area is negligible at a given scale (e.g., water wells, elevation bench marks, light poles).
- **Lines (Arcs):** One-dimensional features consisting of an ordered series of connected coordinates. They possess length but no width (e.g., centerlines of roads, streams, utility networks).
- **Polygons (Areas):** Two-dimensional closed loops enclosed by boundary lines. They possess both perimeter and area properties (e.g., land parcels, administrative boundaries, lakes).

B. Structural Characteristics

- **Object-Based Approach:** Every object is an independent entity with a direct row linked to an attribute database table.
- **Topology:** advanced vector models build explicit topological relationships (defining connectivity, adjacency, and containment), allowing for clean network analysis and preventing boundary overlaps.
- **Best Suited For:** Discrete features with precise, crisp boundaries (e.g., engineering layouts, cadastral parcel maps, transportation networks).

5. Raster Spatial Data Model

The raster data model represents the geographic space as a continuous, regular grid of matrix cells or pixels. Space is not empty; instead, every single location has a value.

A. Structural Elements

- **Grid Cells (Pixels):** The smallest indivisible unit of a raster layer. Each cell contains a single numeric value representing a specific attribute or condition at that location.
- **Cell Size (Resolution):** The ground area covered by a single pixel (e.g., $10\text{ m} \times 10\text{ m}$ resolution). Smaller cell sizes yield higher spatial resolution and sharper details, but exponentially increase storage requirements.
- **Implicit Location:** Unlike vectors, explicit coordinates are not stored for every cell. Only the origin coordinate (usually top-left corner) and cell size are saved; all other positions are computed implicitly based on row and column indexes.

B. Structural Characteristics

- **Field-Based Approach:** Represents phenomena that vary continuously across space without distinct boundaries.
- **Simplicity:** The data structure is a straightforward matrix array, making map algebra, grid overlays, and mathematical modeling computationally fast and highly efficient.

- **Best Suited For:** Continuous surfaces, environmental layers, and imagery (e.g., Digital Elevation Models (DEM), rainfall distribution, temperature gradients, or satellite spectral bands).

Comparison between vector and raster data models

Characteristic	Vector Data Model	Raster Data Model
Data Structure	Highly complex (Coordinates + Topology)	Simple grid matrix (Rows + Columns)
Representation	Discrete points, lines, polygons	Continuous surfaces / pixels
Spatial Precision	Very high; limited only by coordinate accuracy	Limited by the cell resolution size
Analysis Focus	Network analysis, proximity, buffering	Map algebra, continuous surface modeling
File Storage Size	Small/Compact (typically lines/points)	Extremely large (depends on pixel size/extent)
Primary Reference	Heavily used in ArcGIS Manuals & Korte	Emphasized in Environmental Studies (Burrough)